Lecture 14

Schur–Weyl duality

Objectives. *Lecture 13 found the characters of S_n as Schur functions. This last lecture finds the same functions as characters of* $\mathrm{GL}(V)$*, where* $d = \dim V$*. On* $V^{\otimes n}$*,* S_n *permutes the tensor places and* $\mathrm{GL}(V)$ *acts in every place at once; each spans the commutant of the other. So* $V^{\otimes n}$ *splits into pieces* $V^\lambda \otimes \mathbb{S}_\lambda(V)$*.*

By the end of this lecture you will be able to:

- Prove that the image A of $\mathbb{C}[G]$ in $\mathrm{End}(W)$ satisfies $A'' = A$, and that the non-zero multiplicity spaces are irreducible and pairwise non-isomorphic for A'.
- Prove that the operators on $V^{\otimes n}$ commuting with S_n are spanned by the $g^{\otimes n}$, $g \in \mathrm{GL}(V)$.
- Decompose $V^{\otimes n}$ under S_n and $\mathrm{GL}(V)$, and prove that $\mathbb{S}_\lambda(V) = 0$ exactly when $\ell(\lambda) > \dim V$, and that otherwise these modules are irreducible and pairwise non-isomorphic, with $\operatorname{char} \mathbb{S}_\lambda(V) = s_\lambda(x_1, \dots, x_d)$.
- For $\ell(\lambda) \le d$, compute $\dim \mathbb{S}_\lambda(\mathbb{C}^d)$ as the number of Gelfand–Tsetlin patterns with top row λ.

14.1 Commuting actions

Throughout, $n \ge 1$, and V is a complex vector space with a basis $e_1, \dots, e_d$, $d \ge 1$; a subscripted e_i is a basis vector, not the identity e of S_n. The d^n tensors $e_{i_1} \otimes \cdots \otimes e_{i_n}$ are a basis of $V^{\otimes n}$.

Definition 14.1 (The two actions). S_n acts on $V^{\otimes n}$ by permuting the places:

$$\sigma \cdot (v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)},$$

so the factor in place i moves to place $\sigma(i)$; we write σ for the operator too. $\mathrm{GL}(V)$ acts by $g^{\otimes n}$, applying g in every place.

For instance,

$$(1\,2\,3) \cdot (v_1 \otimes v_2 \otimes v_3) = v_3 \otimes v_1 \otimes v_2.$$

The place action is a left action: place i of $\sigma \cdot (\tau \cdot v)$ holds the factor of v from place

$$\tau^{-1}(\sigma^{-1}(i)) = (\sigma\tau)^{-1}(i).$$

A $\mathrm{GL}(V)$-*module* is a finite-dimensional space M with a homomorphism $\mathrm{GL}(V) \to \mathrm{GL}(M)$; submodules, irreducibility and isomorphisms are as in Lecture 5, but $\mathrm{GL}(V)$ is infinite, and nothing is averaged over it.

For any $a \in \mathrm{End}(V)$, write $a^{\otimes n}$ for the operator applying a in every place. The two actions commute: indeed,

$$\begin{aligned}\sigma\, a^{\otimes n}(v_1 \otimes \cdots \otimes v_n) &= \sigma \cdot (av_1 \otimes \cdots \otimes av_n)\\ &= av_{\sigma^{-1}(1)} \otimes \cdots \otimes av_{\sigma^{-1}(n)}\\ &= a^{\otimes n}\,\sigma(v_1 \otimes \cdots \otimes v_n).\end{aligned}$$

So every $g^{\otimes n}$ commutes with the span A of the σ; Corollary 14.8 proves the converses.

14.2 Double centralisers for an image of $\mathbb{C}[G]$

To understand the two commuting actions, we first prove a double centraliser theorem for an image of a group algebra. Here G is a finite group, (ρ, W) a representation, and $A = \rho(\mathbb{C}[G])$, the span of the $\rho(g)$, a subalgebra of $\mathrm{End}(W)$ (Proposition 7.5(i)). For a subalgebra $B \subseteq \mathrm{End}(W)$, the *commutant* is

$$B' = \{S \in \mathrm{End}(W) : Sb = bS \text{ for all } b \in B\},$$

and $B'' = (B')'$. Always $A \subseteq A''$, and $A' = \mathrm{End}_G(W)$.

Put $M_U = \mathrm{Hom}_G(U, W)$, the multiplicity space from Lecture 7. The isomorphism of Proposition 7.14 is $\Theta\colon \bigoplus_{U \in \mathrm{Irr}(G)} U \otimes M_U \to W$, with $u \otimes f \mapsto f(u)$. Through it, $a \in \mathbb{C}[G]$ acts on $U \otimes M_U$ by $\rho_U(a) \otimes \mathbb{1}$. By Theorem 7.17, $\Psi\colon A' \to \bigoplus_U \mathrm{End}(M_U)$, $\Psi(S)_U(f) = S \circ f$, is an isomorphism; $S \in A'$ acts on $U \otimes M_U$ by $\mathbb{1} \otimes \Psi(S)_U$. We say that U *occurs* in W if $M_U \neq 0$.

Lemma 14.2 (A commutant in a tensor product)**.** *Let U and $M \neq 0$ be finite-dimensional complex vector spaces. The operators on $U \otimes M$ that commute with every $\mathbb{1}_U \otimes B$, $B \in \mathrm{End}(M)$, are the $X \otimes \mathbb{1}_M$, $X \in \mathrm{End}(U)$.*

Proof. The $X \otimes \mathbb{1}_M$ commute with them. Conversely, let S commute with them; take a basis $f_1, \ldots, f_m$ of M, the matrix units E_{ab} ($E_{ab} f_b = f_a$, and $E_{ab} f_k = 0$ for $k \neq b$), and write

$$S(u \otimes f_j) = \sum_k S_{kj}(u) \otimes f_k,$$

with $S_{kj} \in \mathrm{End}(U)$. Apply $S(\mathbb{1} \otimes E_{ab}) = (\mathbb{1} \otimes E_{ab})S$ to $u \otimes f_b$:

$$\sum_k S_{ka}(u) \otimes f_k = S_{bb}(u) \otimes f_a.$$

So $S_{ka} = 0$ for $k \neq a$, and $S_{aa} = S_{bb}$ for all a, b: $S = S_{11} \otimes \mathbb{1}_M$. ∎

Theorem 14.3 (The double centraliser)**.** *Let W be a representation of G, and A the image of $\mathbb{C}[G]$ in $\mathrm{End}(W)$.*

(i) Through Θ, A consists of the operators acting on each $U \otimes M_U$ by $X_U \otimes \mathbb{1}$, with $X_U \in \mathrm{End}(U)$ arbitrary for the U that occur. So $\dim A = \sum (\dim U)^2$ over the U that occur.

(ii) $A'' = A$.

(iii) A' acts on each M_U by $S \cdot f = S \circ f$, and Θ intertwines this action. The non-zero M_U are irreducible and pairwise non-isomorphic A'-modules.

Proof. (i) By Theorem 7.9, as a runs over $\mathbb{C}[G]$, the tuple $(\rho_U(a))_U$ runs over every tuple $(X_U)_U$. The summands with $M_U = 0$ are zero, and $X_U \otimes \mathbb{1}$ determines X_U when $M_U \neq 0$.

(ii) Let $S \in A''$. *The blocks.* As Ψ is onto, A' contains the π_U with $\Psi(\pi_U)$ the identity on M_U and 0 on the other $M_{U'}$: through Θ, the projection onto $U \otimes M_U$. As S commutes with π_U, it maps $U \otimes M_U$ into itself.

One block. Let U occur. As Ψ is onto, S commutes on $U \otimes M_U$ with every $\mathbb{1} \otimes B$, $B \in \operatorname{End}(M_U)$. By Lemma 14.2, S acts there by some $X_U \otimes \mathbb{1}$, so the block description in (i) gives $S \in A$.

(iii) Θ intertwines, as noted before Lemma 14.2. As Ψ is onto, A' acts on M_U through all of $\operatorname{End}(M_U)$, which takes any $f \neq 0$ to any vector. And π_U acts on M_U by the identity, on $M_{U'}$ by 0.

So an isomorphism $\varphi\colon M_U \to M_{U'}$ of A'-modules, $U' \neq U$, would give

$$\varphi(f) = \varphi(\pi_U \cdot f) = \pi_U \cdot \varphi(f) = 0$$

for every f, impossible if $M_U \neq 0$. ∎

Example 14.4 (Guard: the image, not $\mathbb{C}[S_n]$)**.** Let S_n act on $V^{\otimes n}$ with $n > d$, and recall the central idempotent (Proposition 7.12) $e_{\mathrm{sgn}} = \frac{1}{n!}\sum_\tau \operatorname{sgn}(\tau)\,\tau$. As $\sigma e_{\mathrm{sgn}} = \operatorname{sgn}(\sigma) e_{\mathrm{sgn}}$, each $y = e_{\mathrm{sgn}} \cdot v$ is *antisymmetric*: $\sigma \cdot y = \operatorname{sgn}(\sigma)\, y$. Each basis tensor has equal factors in two places $a \neq b$, so $(a\,b)$ fixes it while negating y: its coefficient in y is 0. So e_{sgn}, which is non-zero in $\mathbb{C}[S_n]$, acts as 0 on $V^{\otimes n}$: the image A is smaller than $\mathbb{C}[S_n]$.

14.3 Polarisation

We now identify the commutant of the place action. Polarisation recovers symmetric multilinear information from diagonal tensors $a^{\otimes n}$.

Lemma 14.5 (Operators on a tensor power)**.** *The linear map* $\operatorname{End}(V)^{\otimes n} \to \operatorname{End}(V^{\otimes n})$ *taking* $a_1 \otimes \cdots \otimes a_n$ *to* $v_1 \otimes \cdots \otimes v_n \mapsto a_1 v_1 \otimes \cdots \otimes a_n v_n$ *is an isomorphism, and conjugation by* σ *is the place action:*

$$\sigma\,(a_1 \otimes \cdots \otimes a_n)\,\sigma^{-1} = a_{\sigma^{-1}(1)} \otimes \cdots \otimes a_{\sigma^{-1}(n)}.$$

Proof. With the matrix units $E_{ij}e_k = \delta_{jk}e_i$ of $\operatorname{End}(V)$, the map sends $E_{i_1 j_1} \otimes \cdots \otimes E_{i_n j_n}$ to the matrix unit of $V^{\otimes n}$ taking $e_{j_1} \otimes \cdots \otimes e_{j_n}$ to $e_{i_1} \otimes \cdots \otimes e_{i_n}$: a basis to a basis. For conjugation, put $v = v_1 \otimes \cdots \otimes v_n$:

$$\begin{aligned}\sigma(a_1 \otimes \cdots \otimes a_n)\sigma^{-1} v &= \sigma(a_1 \otimes \cdots \otimes a_n)(v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)}) \\ &= \sigma \cdot \left(a_1 v_{\sigma(1)} \otimes \cdots \otimes a_n v_{\sigma(n)}\right) \\ &= a_{\sigma^{-1}(1)} v_1 \otimes \cdots \otimes a_{\sigma^{-1}(n)} v_n.\end{aligned}$$ ∎

Lemma 14.6 (Polarisation)**.** *The invariants for the place action of* S_n *on* $\operatorname{End}(V)^{\otimes n}$*, defined as for* $V^{\otimes n}$*, are spanned by the tensors* $a^{\otimes n}$*,* $a \in \operatorname{End}(V)$*.*

Proof. *Averaging.* Each $a^{\otimes n}$ is invariant. The average $P = \frac{1}{n!}\sum_\sigma \sigma$ projects onto the invariants (Proposition 5.8(ii)), so they are spanned by the $P(a_1 \otimes \cdots \otimes a_n)$.

The coefficient. For $t \in \mathbb{C}^n$, by multilinearity,

$$(t_1 a_1 + \cdots + t_n a_n)^{\otimes n} = \sum_f t_{f(1)} \cdots t_{f(n)}\, a_{f(1)} \otimes \cdots \otimes a_{f(n)},$$

over all maps f from $\{1,\dots,n\}$ to itself. The bijections give $t_1\cdots t_n$; no other map does, since a non-bijection omits an index. The coefficient is therefore $n!\,P(a_1\otimes\cdots\otimes a_n)$.

Spanning. Let η be a linear functional vanishing on every $a^{\otimes n}$. Then $\eta\big((\sum_i t_i a_i)^{\otimes n}\big)$ is a polynomial in t that vanishes on $\mathbb{C}^n$, so its coefficients vanish (by induction on the number of variables), and η vanishes on the invariants. A vector outside the span of the $a^{\otimes n}$ can be added to a basis of that span and the result extended to a basis of $\mathrm{End}(V)^{\otimes n}$. A linear functional which is 1 on that vector and 0 on the other basis vectors then separates it from the span. Hence the invariants lie in the span. ∎

Theorem 14.7 (The commutant of S_n)**.** $\mathrm{End}_{S_n}(V^{\otimes n})$ *is the span of the* $g^{\otimes n}$, $g\in\mathrm{GL}(V)$.

Proof. An operator T commutes with every σ exactly when $\sigma T\sigma^{-1}=T$. Thus Lemma 14.5 identifies the commutant with the place-action invariants. Under this identification, the tensor $a^{\otimes n}$ becomes the operator $a^{\otimes n}$, so Lemma 14.6 shows that these operators span the commutant. Fix a and a linear functional η vanishing on every $g^{\otimes n}$. The operator $a+t\mathbb{1}$ is invertible unless $-t$ is one of the at most d eigenvalues of a. So $q(t)=\eta\big((a+t\mathbb{1})^{\otimes n}\big)$ is a polynomial vanishing at all those other t. It has infinitely many zeros, so it is the zero polynomial and

$$\eta(a^{\otimes n})=q(0)=0.$$

The separating-functional argument of Lemma 14.6 puts $a^{\otimes n}$ in the span of the $g^{\otimes n}$. Conversely, each $g^{\otimes n}$ commutes with every σ (§ 14.1). ∎

Corollary 14.8 (Each action spans the other's commutant)**.** *Let* A *be the image of* $\mathbb{C}[S_n]$ *in* $\mathrm{End}(V^{\otimes n})$*. Then* A' *is the span of the* $g^{\otimes n}$*, and the operators commuting with every* $g^{\otimes n}$ *form* A*.*

Proof. $A'=\mathrm{End}_{S_n}(V^{\otimes n})$ is the span by Theorem 14.7; so the operators commuting with every $g^{\otimes n}$ form $A''=A$ (Theorem 14.3(ii)). ∎

14.4 The decomposition and its characters

From here on, $x=(x_1,\dots,x_d)$.

Definition 14.9 (Schur modules)**.** For $\lambda\vdash n$, the *Schur module* is the multiplicity space

$$\mathbb{S}_\lambda(V)=\mathrm{Hom}_{S_n}(V^\lambda,V^{\otimes n}),$$

with $g\cdot f=g^{\otimes n}\circ f$, an intertwiner as $g^{\otimes n}$ commutes with S_n. This is a $\mathrm{GL}(V)$-action since $(gh)^{\otimes n}=g^{\otimes n}h^{\otimes n}$. For x with every $x_i\neq 0$, its *character* $\mathrm{char}\,\mathbb{S}_\lambda(V)(x)$ is the trace of the induced action of the $g\in\mathrm{GL}(V)$ whose matrix on V in the chosen basis is $\mathrm{diag}(x)$.

As $V^{(1^n)}=\mathrm{sgn}$ (§ 12.2), $f\mapsto f(u)$, for u spanning sgn, identifies $\mathbb{S}_{(1^n)}(V)$ with the antisymmetric tensors in $V^{\otimes n}$, in turn identified with the exterior power $\Lambda^n(V)$ (with a space, unlike Lecture 13's Λ^n).

Lemma 14.10 (The trace)**.** *For* $g=\mathrm{diag}(x)$ *and* $\sigma\in S_n$ *of type* μ*,* $\mathrm{tr}(\sigma g^{\otimes n})=p_\mu(x)$.

Proof. $\sigma g^{\otimes n}$ takes each basis tensor to a multiple of another. The latter is the same tensor exactly when its index $(i_1,\dots,i_n)$ is constant on each cycle c of σ; the multiple is then

$x_{i_1} \cdots x_{i_n}$. So

$$\operatorname{tr}(\sigma g^{\otimes n}) = \prod_c \left(x_1^{|c|} + \cdots + x_d^{|c|}\right) = p_\mu(x),$$

with $|c|$ the length of c (as for Theorem 13.5(ii)). ∎

Lemma 14.11 (Stability). *Let $N \geq d$ and $\ell(\lambda) \leq N$. Setting $x_{d+1}, \ldots, x_N$ to 0 sends s_λ in N variables to $s_\lambda(x)$ if $\ell(\lambda) \leq d$, and to 0 otherwise.*

Proof. For $d \leq M < N$, consider setting the last of $M+1$ variables to 0, with $\ell(\lambda) \leq M+1$. With $\alpha = \lambda + \delta$, the last column of $a_\alpha = \det(x_j^{\alpha_i})$ becomes the 0^{α_i} ($0^0 = 1$).

If $\ell(\lambda) \leq M$, it is the last unit vector: $\alpha_{M+1} = 0$ and $\alpha_i \geq 1$ for $i \leq M$. Expand along it, and take x_j out of column j: as $\alpha_i - 1 = \lambda_i + M - i$,

$$a_{\lambda+\delta}(x_1, \ldots, x_M, 0) = x_1 \cdots x_M \, a_{\lambda+\delta}(x_1, \ldots, x_M),$$

with δ taken in M variables on the right; likewise for a_δ. Substitute in $a_{\lambda+\delta} = a_\delta s_\lambda$ and cancel the non-zero polynomial $x_1 \cdots x_M a_\delta(x_1, \ldots, x_M)$ to get s_λ in M variables.

If $\ell(\lambda) = M+1$, every $\alpha_i \geq 1$, the last column is 0, and $a_\delta(x_1, \ldots, x_M, 0)\, s_\lambda(x_1, \ldots, x_M, 0) = 0$. The first factor is non-zero by the first case with $\lambda = \varnothing$, so the specialised Schur polynomial is 0. Iterating from N variables down to d, a shape of length greater than d reaches the second case at $M = \ell(\lambda) - 1$; subsequent substitutions leave 0 unchanged. ∎

In $\max(n, d)$ variables, $p_\mu = \sum_{\lambda \vdash n} \chi^\lambda(\mu)\, s_\lambda$ (Corollary 13.17), so by Lemma 14.11,

$$p_\mu(x) = \sum_{\lambda \vdash n,\ \ell(\lambda) \leq d} \chi^\lambda(\mu)\, s_\lambda(x). \tag{$*$}$$

Theorem 14.12 (Schur–Weyl duality).

(i) *$u \otimes f \mapsto f(u)$ is an isomorphism*

$$\bigoplus_{\lambda \vdash n,\ \ell(\lambda) \leq d} V^\lambda \otimes \mathbb{S}_\lambda(V) \to V^{\otimes n},$$

through which σ acts by $\sigma \otimes \mathbb{1}$ and $g^{\otimes n}$ by $\mathbb{1} \otimes g$.

(ii) *For $\ell(\lambda) \leq d$, $\mathbb{S}_\lambda(V)$ is a non-zero irreducible $\mathrm{GL}(V)$-module, and $\mathbb{S}_\lambda(V) \ncong \mathbb{S}_\nu(V)$ for $\nu \neq \lambda$. For $\ell(\lambda) > d$, $\mathbb{S}_\lambda(V) = 0$.*

(iii) *For $\ell(\lambda) \leq d$ and every $x \in (\mathbb{C}^\times)^d$, $\operatorname{char} \mathbb{S}_\lambda(V)(x) = s_\lambda(x)$.*

Proof. *The trace.* Over all $\lambda \vdash n$, the map is Proposition 7.14's Θ, and $g^{\otimes n} \in A'$ acts by $\mathbb{1} \otimes g$ (Theorem 14.3(iii)). For $g = \operatorname{diag}(x)$, every $x_i \neq 0$, as $\operatorname{tr}(X \otimes Y) = \operatorname{tr} X \operatorname{tr} Y$,

$$\operatorname{tr}(\sigma g^{\otimes n}) = \sum_{\lambda \vdash n} \chi^\lambda(\sigma)\, \operatorname{char} \mathbb{S}_\lambda(V)(x).$$

By Lemma 14.10 and $(*)$ it is also $\sum_{\ell(\lambda) \leq d} \chi^\lambda(\sigma)\, s_\lambda(x)$. The χ^λ are linearly independent (Corollary 6.8(v)), so $\operatorname{char} \mathbb{S}_\lambda(V)(x)$ is $s_\lambda(x)$ if $\ell(\lambda) \leq d$, and 0 otherwise.

Which occur. At $x = (1, \ldots, 1)$ the character is the dimension, so $\mathbb{S}_\lambda(V) = 0$ if $\ell(\lambda) > d$. If $\ell(\lambda) \leq d$, the exponents of $x^{\lambda+\delta}$ are strictly decreasing, so this monomial has coefficient 1 in $a_{\lambda+\delta}$. Since $a_{\lambda+\delta} = a_\delta s_\lambda$, the polynomial s_λ is non-zero. The polynomial $x_1 \cdots x_d\, s_\lambda(x)$ is then non-zero at some $x \in \mathbb{C}^d$; each $x_i \neq 0$ and $s_\lambda(x) \neq 0$ there. The character equals $s_\lambda(x)$ at this point, so $\mathbb{S}_\lambda(V) \neq 0$. The zero summands can therefore be omitted from Θ in (i).

Irreducible and distinct. By Theorem 14.3(iii), the non-zero $\mathbb{S}_\lambda(V)$ are irreducible and pairwise non-isomorphic for A', the span of the $g^{\otimes n}$ (Theorem 14.7). A subspace stable under every $g^{\otimes n}$ is stable under their span, and a map intertwining all of them intertwines their span. Thus the modules are irreducible and pairwise non-isomorphic for $\mathrm{GL}(V)$. ∎

Example 14.13 (Guard: polynomial modules only). The action $g \cdot z = \det(g)^{-1} z$ on $\mathbb{C}$ has character $(x_1 \cdots x_d)^{-1}$, not a polynomial: it is no $\mathbb{S}_\lambda(V)$. Such modules belong to Lie theory.

14.5 Interlacing and Gelfand–Tsetlin patterns

We can compute the dimensions of Schur modules by counting patterns. For $d \geq 2$ and partitions with $\ell(\mu) < d$ and $\ell(\lambda) \leq d$, padded, μ *interlaces* λ, $\mu \prec \lambda$, if

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \mu_{d-1} \geq \lambda_d.$$

The same definition applies to rows of lengths $k-1$ and k, with k in place of d.

Theorem 14.14 (Branching). *Let $d \geq 2$ and $\ell(\lambda) \leq d$. With $y = (x_1, \ldots, x_{d-1})$,*

$$s_\lambda(y,t) = \sum_{\mu \prec \lambda} s_\mu(y)\, t^{|\lambda|-|\mu|}.$$

Proof. Let $\alpha = \lambda + \delta$, of length d.

Row operations. For $i = 1, \ldots, d-1$ in turn, subtract $t^{\alpha_i - \alpha_{i+1}}$ times row $i+1$ from row i of $a_\alpha(y,t)$, which keeps the determinant. The last column becomes $(0, \ldots, 0, t^{\alpha_d})$, so

$$a_\alpha(y,t) = t^{\alpha_d} \det\bigl(x_j^{\alpha_i} - t^{\alpha_i - \alpha_{i+1}} x_j^{\alpha_{i+1}}\bigr)_{i,j<d}.$$

Geometric sums. Each entry is

$$(x_j - t) \sum_{\beta_i = \alpha_{i+1}}^{\alpha_i - 1} x_j^{\beta_i}\, t^{\alpha_i - 1 - \beta_i}.$$

Take $x_j - t$ out of column j, and expand each row by linearity:

$$a_\alpha(y,t) = \prod_{j<d} (x_j - t) \sum_{\beta} t^{\alpha_d + \sum_i (\alpha_i - 1 - \beta_i)}\, a_\beta(y),$$

over the β with $\alpha_{i+1} \leq \beta_i < \alpha_i$.

Interlacing. Put $\beta = \mu + \delta$, of length $d-1$, so that

$$\alpha_{i+1} \leq \beta_i < \alpha_i \iff \lambda_{i+1} \leq \mu_i \leq \lambda_i :$$

the β are the $\mu \prec \lambda$, with t to the power

$$\begin{aligned} \alpha_d + \sum_{i<d} (\alpha_i - 1 - \beta_i) &= \lambda_d + \sum_{i<d} (\lambda_i - \mu_i) \\ &= |\lambda| - |\mu|\,. \end{aligned}$$

By Lemma 13.7(ii), $a_\delta(y,t) = a_\delta(y) \prod_{j<d} (x_j - t)$. Divide. ∎

Definition 14.15 (Gelfand–Tsetlin patterns)**.** A *Gelfand–Tsetlin pattern* with top row λ, $\ell(\lambda) \le d$, has d rows of non-negative integers, row k from the bottom with entries $\lambda_{k,1}, \dots, \lambda_{k,k}$; the top row is λ, padded, and $\lambda_{k,i} \ge \lambda_{k-1,i} \ge \lambda_{k,i+1}$ for $2 \le k \le d$ and $1 \le i \le k-1$. Thus row $k-1$ interlaces row k. Its *weight* w has w_k the sum of row k minus that of row $k-1$ (row 0 is empty).

Corollary 14.16 (The pattern count)**.** *For $\ell(\lambda) \le d$, $s_\lambda(x) = \sum_P x_1^{w_1} \cdots x_d^{w_d}$, over the patterns P with top row λ, w the weight of P. So $\dim \mathbb{S}_\lambda(V)$ is the number of these patterns.*

Proof. Base. For $d = 1$, $s_{(m)}(x_1) = x_1^m$, from one pattern.

Induction. For $d \ge 2$, the rows below the top form a pattern with top row some $\mu \prec \lambda$, and $w_d = |\lambda| - |\mu|$: apply Theorem 14.14 with $t = x_d$.

Dimension. Set $x = (1, \dots, 1)$ (Theorem 14.12(iii)). ∎

Theorem 14.14's module form (from GL_d to GL_{d-1}) and a basis indexed by the patterns are Lie theory's.

14.6 Worked example

The decomposition. Let $V = \mathbb{C}^2$, $n = 3$. The partitions of 3 with at most two rows are (3) and $(2,1)$, so

$$(\mathbb{C}^2)^{\otimes 3} \cong V^{(3)} \otimes \mathbb{S}_{(3)}(\mathbb{C}^2) \oplus V^{(2,1)} \otimes \mathbb{S}_{(2,1)}(\mathbb{C}^2),$$

and $\mathbb{S}_{(1^3)}(\mathbb{C}^2)$, which is $\Lambda^3(\mathbb{C}^2)$, is 0.

Characters. A pattern with top row $(3,0)$ has one bottom entry b, $3 \ge b \ge 0$, and weight $(b, 3-b)$. Those with top row $(2,1)$ are

$$\begin{array}{ccc} 2 & & 1 \\ & 2 & \end{array} \text{ of weight } (2,1), \qquad \begin{array}{ccc} 2 & & 1 \\ & 1 & \end{array} \text{ of weight } (1,2).$$

By Corollary 14.16,

$$\begin{aligned} s_{(3)}(x_1, x_2) &= x_1^3 + x_1^2 x_2 + x_1 x_2^2 + x_2^3, \\ s_{(2,1)}(x_1, x_2) &= x_1^2 x_2 + x_1 x_2^2. \end{aligned}$$

Dimensions. At $x = (1,1)$,

$$\begin{aligned} \dim(\mathbb{C}^2)^{\otimes 3} &= f^{(3)}\, s_{(3)}(1,1) + f^{(2,1)}\, s_{(2,1)}(1,1) \\ &= 1 \cdot 4 + 2 \cdot 2 = 8. \end{aligned}$$

The two algebras. The double-centraliser and commutant formulas (Theorem 14.3, Theorem 7.17) give

$$\dim A = 1^2 + 2^2 = 5, \qquad \dim A' = 4^2 + 2^2 = 20.$$

Example 14.17 (Guard: V^λ is not $\mathbb{S}_\lambda(V)$)**.** The representation $V^{(3)} = \mathbf{1}$ has dimension 1, but $\mathbb{S}_{(3)}(\mathbb{C}^2)$ has dimension 4. Generally, V^λ is an S_n-module of dimension f^λ; the $\mathrm{GL}(V)$-module $\mathbb{S}_\lambda(V)$ has dimension depending on d.

Self-check. Decompose $(\mathbb{C}^3)^{\otimes 3}$. *Answer:* the three summands in Schur–Weyl duality are

$$\big(V^{(3)} \otimes \mathbb{S}_{(3)}(\mathbb{C}^3)\big) \oplus \big(V^{(2,1)} \otimes \mathbb{S}_{(2,1)}(\mathbb{C}^3)\big) \oplus \big(V^{(1^3)} \otimes \mathbb{S}_{(1^3)}(\mathbb{C}^3)\big).$$

To count patterns, pad each top row to length three. For $(3,0,0)$, the middle row is $(a,0)$ with $0 \le a \le 3$; there are $a+1$ choices for the bottom entry, giving $1+2+3+4=10$. For $(2,1,0)$, the middle rows $(2,1),(2,0),(1,1),(1,0)$ have respectively $2,3,1,2$ choices for the bottom entry, giving 8. For $(1,1,1)$ there is just one pattern. Thus

λ	(3)	$(2,1)$	(1^3)
$\dim \mathbb{S}_\lambda(\mathbb{C}^3)$	10	8	1

Consequently,

$$\dim(\mathbb{C}^3)^{\otimes 3} = 1\cdot 10 + 2\cdot 8 + 1\cdot 1 = 27.$$

Summary.

- *The main results:* the double centraliser theorem (Theorem 14.3); the $g^{\otimes n}$, $g \in \mathrm{GL}(V)$, span the commutant of S_n (Theorem 14.7); and Schur–Weyl duality decomposes $V^{\otimes n}$ as $\bigoplus_{\lambda \vdash n,\, \ell(\lambda)\le d} V^\lambda \otimes \mathbb{S}_\lambda(V)$. For $\ell(\lambda) \le d$, these modules are irreducible and pairwise non-isomorphic, with $\operatorname{char} \mathbb{S}_\lambda(V) = s_\lambda$; for greater length the module is zero (Theorem 14.12). Branching and the pattern count are Theorem 14.14 and Corollary 14.16.
- *Handed on* to *Lie Theory*: these, with the Gelfand–Tsetlin basis and Young's orthogonal form of Lectures 10 to 12.

References

Etingof et al., *Introduction to Representation Theory*: Thm 5.18.1, p. 120; Thm 5.18.4, p. 121; Prop. 5.19.1 and Cor. 5.19.2, p. 122; Thm 5.22.1, p. 129 (their L_λ is our $\mathbb{S}_\lambda(V)$). Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Thm 8.1.3, p. 364; Ex. 8.1.4, p. 365; Thms 8.2.8 and 8.2.10, pp. 375–378. Fulton and Harris, *Representation Theory*: Thm 6.3, pp. 77–78, and Lemma 6.23, p. 86.[1] Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed.: stability, p. 41; branching, (5.11), p. 72, via skew Schur functions.

[1] S_d acts on the right (p. 76): their $\nu \cdot \sigma$ is our $\sigma^{-1} \cdot \nu$.